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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2025

Fourth, Seventh Semester

21CSE253T - INTERNET OF THINGS

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

- Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- Part - B and Part - C should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Marks CO BL

Answer all Questions

- | | | | |
|---|---|---|---|
| 1. Which layer in the simplified IoT functional stack is responsible for sensing and initial data collection? | 1 | 1 | 1 |
| (A) Perception (or sensing) layer | | | |
| (B) Application layer | | | |
| (C) Network layer | | | |
| (D) Business layer | | | |
| 2. Which of the following best describes Fog computing in IoT? | 1 | 1 | 2 |
| (A) Centralized processing only in remote cloud datacenters | | | |
| (B) Decentralized processing and storage closer to edge devices to reduce latency | | | |
| (C) A hardware-only solution replacing cloud services | | | |
| (D) A proprietary protocol for device identity | | | |
| 3. Which device is typically classified as an actuator (not a sensor)? | 1 | 1 | 2 |
| (A) Temperature sensor | | | |
| (B) Humidity sensor | | | |
| (C) Servo motor that opens a valve | | | |
| (D) Ambient light sensor | | | |
| 4. Which computing reduces delay by processing data near the source? | 1 | 1 | 2 |
| (A) Cloud | | | |
| (B) Edge | | | |
| (C) Virtual | | | |
| (D) Remote | | | |
| 5. LoRaWAN is best suited for which scenario? | 1 | 2 | 2 |
| (A) High-bandwidth video streaming | | | |
| (B) Real-time control requiring microsecond latency | | | |
| (C) Short-range, high-throughput LAN | | | |
| (D) Long-range, low-power sensor telemetry | | | |
| 6. 6LoWPAN's main purpose is to: | 1 | 2 | 2 |
| (A) Replace TCP with a new transport protocol | | | |
| (B) Provide encryption for wireless sensor data | | | |
| (C) Compress and adapt IPv6 for low-power lossy networks | | | |
| (D) Act as a routing protocol for Wi-Fi networks | | | |
| 7. MQTT uses which messaging pattern? | 1 | 2 | 1 |
| (A) Request/response | | | |
| (B) Peer-to-peer file sharing | | | |
| (C) Broadcast only | | | |
| (D) Publish/subscribe | | | |
| 8. Which protocol is RESTful, lightweight and designed for constrained devices? | 1 | 2 | 1 |
| (A) CoAP | | | |
| (B) FTP | | | |
| (C) SNMP | | | |
| (D) SMTP | | | |

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|--|--|---|---|
| 9. Which component in a microcontroller executes instructions? | 1 | 3 | 1 |
| (A) ADC | (B) CPU (core) | | |
| (C) GPIO header | (D) UART peripheral | | |
| 10. Arduino sketches are written in a language based on: | 1 | 3 | 1 |
| (A) Python | (B) C / C++ | | |
| (C) JavaScript | (D) Ruby | | |
| 11. A System on Chip (SoC) differs from a microcontroller mainly because: | 1 | 3 | 2 |
| (A) SoC integrates CPU, memory, and many peripherals (sometimes GPUs, radios) on a single silicon die, targeting higher integration and performance. | (B) Microcontrollers always contain GPUs. | | |
| (C) SoCs are only used in cloud servers | (D) Microcontrollers always include an OS | | |
| 12. Which bus is commonly used for communication with many small sensors on the same two-wire bus in IoT prototypes? | 1 | 3 | 1 |
| (A) HDMI | (B) I ² C | | |
| (C) SATA | (D) VGA | | |
| 13. Which database is a document-oriented NoSQL store commonly used in IoT? | 1 | 4 | 1 |
| (A) MySQL | (B) MongoDB | | |
| (C) Oracle RDBMS | (D) MS Access | | |
| 14. Apache Kafka is primarily used for: | 1 | 4 | 2 |
| (A) Distributed streaming and event buffering | (B) Relational transactions | | |
| (C) Model training only | (D) Front-end UI rendering | | |
| 15. Edge analytics in IoT mainly helps to: | 1 | 4 | 2 |
| (A) Reduce the amount of data sent to the cloud | (B) Increase the number of sensors required | | |
| (C) Lower device battery capacity | (D) Replace all cloud services | | |
| 16. NETCONF uses which modeling language for network device configuration? | 1 | 4 | 1 |
| (A) SQL | (B) YANG | | |
| (C) HTML | (D) Protobuf | | |
| 17. IBM Watson IoT Platform primarily offers: | 1 | 5 | 1 |
| (A) On-device encryption chips | (B) Manufacture of sensors and actuators | | |
| (C) Local PLC firmware only | (D) Cloud device management and analytics services | | |
| 18. CPwE stands for: | 1 | 5 | 1 |
| (A) Converged Plantwide Ethernet | (B) Centralized Plant Web Engine | | |
| (C) Cloud-Provided Wireless Ethernet | (D) Control Protocol for Ethernet | | |
| 19. Which sensor type is commonly used for detecting vehicle presence in smart parking? | 1 | 5 | 2 |
| (A) Magnetic sensor | (B) Thermocouple | | |
| (C) Light bulb | (D) Barcode scanner | | |

20. The GridBlocks reference model belongs to which domain?	1	5	1
(A) Healthcare records			
(B) Web application frameworks			
(C) Smart grid / power systems			
(D) Agricultural sensors			

PART - B (5 × 8 = 40 Marks)

Marks CO BL

Answer **all** Questions

21. a) Explain the evolution of IoT from M2M and describe the key enabling technologies	8	1	2
(OR)			
b) Describe sensors, actuators and smart objects and explain how smart objects are connected in an IoT ecosystem (include simple examples)	8	1	2
22. a) Compare MQTT and CoAP with respect to (i.e., in terms of) their architecture, resource usage, and suitable use-cases	8	2	4
(OR)			
b) Analyze how 6LoWPAN adapts IPv6 for low-power wireless networks	8	2	4
23. a) Describe an IoT design methodology (requirements → deployment) with key steps and checks.	8	3	3
(OR)			
b) Distinguish the roles of Arduino and Raspberry Pi in an IoT course lab by examining their architectures, programming approaches, power consumption, and use-case suitability.	8	3	3
24. a) Explain how Hadoop, Kafka and Spark fit in an IoT data pipeline; give a short example pipeline.	8	4	3
(OR)			
b) Describe NoSQL databases commonly used in IoT (document stores, time-series DBs, key-value stores). Explain when to prefer NoSQL over relational DBs and give examples.	8	4	1
25. a) Explain the typical architecture of a Smart Parking system and list key benefits	8	5	2
(OR)			
b) Describe the layered architecture of Smart & Connected Cities and give one practical example (smart lighting).	8	5	2

PART - C (1 × 15 = 15 Marks)

Marks CO BL

Answer **any 1** Question

26. A smart farming project uses soil-moisture and weather sensors to guide irrigation. The farm requires quick local decisions for watering, limited internet access in fields, and long-term analysis of crop data. Explain the layered IoT architecture and describe how Edge, Fog, and Cloud computing support these tasks.	15	1	3
27. A city hospital network plans a system where wearable sensors track patients' vital signs and city-wide smart sensors monitor traffic, air quality, and streetlights. The hospital needs instant health alerts and secure record storage, while the city requires real-time traffic control and long-term pollution analysis. Explain how IoT applications can address these healthcare and smart city needs with suitable examples.	15	5	3

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